Comparative Analysis of CNN and Vision Transformer Architectures for Plant Trait Prediction

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Abstract

In this report, we explore the use of Vision Transformers (ViT) for plant trait prediction, comparing them with Convolutional Neural Networks (CNNs). We demonstrate the process of transforming target values, pre-processing data, and the architecture used to achieve better performance metrics. The final model structure is presented along with a comparison of its performance against other models.

1 Introduction

In this work, we aim to predict plant traits from photographs, supplemented with ancillary information related to the local climate, soil conditions, and satellite-derived data, using Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs). Our training set comprises 43,363 plant images and corresponding ancillary data gathered from the iNaturalist Database. The six target plant traits: Stem Specific Density, Leaf Area per Leaf Dry Mass, Plant Height, Seed Dry Mass, Leaf Nitrogen, and Leaf Area, are collected from the TRY Database and are denoted as X4, X11, X18, X26, X50, and X3112, respectively.

Traditionally, the task of predicting plant traits from images has been dominated by CNNs, which have demonstrated efficiency in extracting features from raw image data (Schiller et al. 2021). However, CNNs face limitations in achieving high R^2 scores and are often prone to overfitting, particularly in complex, high-dimensional datasets like ours. To address these challenges, this report explores the efficacy of ViTs as an alternative architecture for this task, comparing their performance against various CNN models and regression techniques.

2 Related Works

The preprocessing pipeline closely follows the methodology described in the original paper (Schiller et al. 2021). Specifically:

1. Target values were normalized between 0 and 1, after removing outliers (those that are 3 standard deviations away from the mean). This normalization is crucial for simultaneously training multiple target features, ensuring consistency in magnitude across features. The log10 transformation (Schiller et al. 2021) was removed due to the extra instability in the R^2 score introduced during the inference stage.
2. A quantile transformation was applied to the ancillary data instead of simple normalization, resulting in normally distributed ancillary data. This approach yielded lower target loss, as the ancillary data now followed a normal distribution. However, this transformation was not applied to the target data, as the inverse quantile process introduces instability in the target loss post-inversion.

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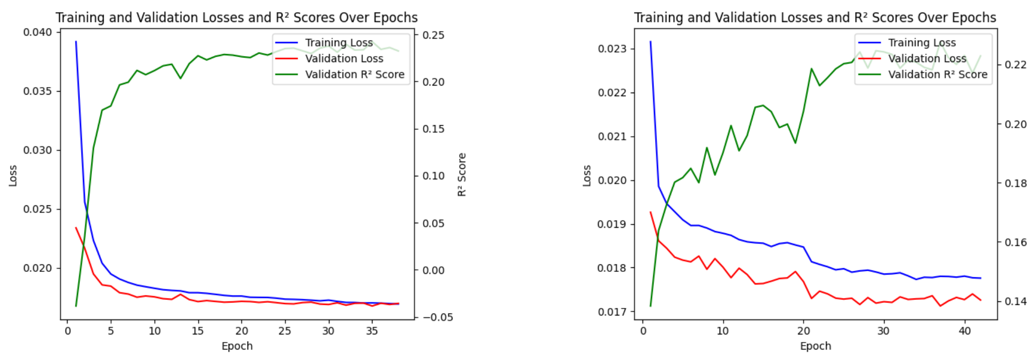
3 Main Results

3.1 Tested Architectures

To replicate the results from the original paper, a CNN + Regression approach was employed (Schiller et al. 2021). Various CNN models, including DenseNet201, MobileNetV2, ResNet18, and ResNet50, were tested. The CNN output, after processing the image, was concatenated with the ancillary data and fed into a regression neural network comprising 3-4 layers. However, it was observed that the CNN models struggled to achieve an R^2 score above 0.3. Simple CNN architectures lacked the capacity to minimize loss effectively, while more complex CNN models tended to overfit, even with the application of careful data augmentation techniques.

Figure 1: Training Loss/R^2 Scores for ResNet50 and DenseNet201 Based Models Over Epochs



(a) ResNet50: Training Loss/R^2 vs. Epochs (b) DenseNet201: Training Loss/R^2 vs. Epochs

To address the limitations of CNNs, Vision Transformers (ViTs) were employed. These models were pre-trained and used as feature encoders for the images, similar to the role played by CNNs. Various transformer models were tested on the first target feature, 'X4_mean,' using a consistent ensembled regression model. The results are summarized in Table 1, where 10% of the training set is used as a validation set.

Table 1: R^2 Scores of Different Vision Transformer & CNN Models on 'X4_mean'

Model	ViT	SWin	DeiT	DINo	ResNet50	ResNet18
R^2 Score	0.4389	0.3577	0.3364	0.3361	0.3015	0.2787

Notably, ViT showed superior performance in terms of loss under the same regression model. The ensembled regression model, composed of several algorithms, also demonstrated varying levels of effectiveness, as shown below:

Table 2: Ensembled Regression Sub-Model Results for X4_mean

Model	KNeighborsDist	CatBoost	KNeighborsUnif	NeuralNetTorch	LightGBMXT	LightGBMLarge
R^2 Score	0.3786	0.3747	0.3737	0.3698	0.3696	0.3667

Model	LightGBM	NeuralNetFastAI	XGBoost	ExtraTreesMSE	RandomForestMSE
R^2 Score	0.3601	0.3511	0.3428	0.3010	0.2960

3.2 Main Structure

The distribution of the target features post-preprocessing is illustrated in Figure 2. The observed differences in distributions suggest that each target feature should be trained independently to minimize loss effectively.

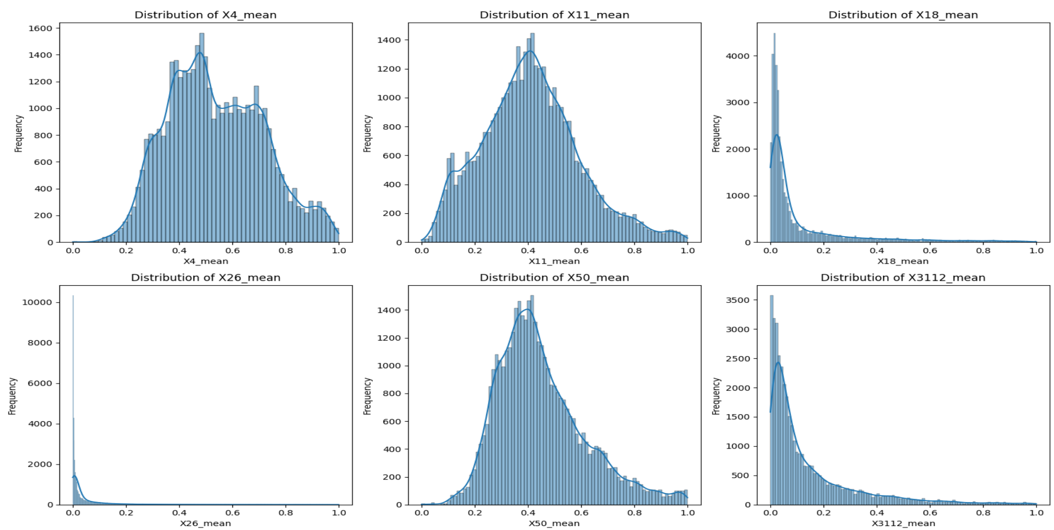


Figure 2: Distribution of Target Features After Preprocessing.

The final model architecture combines a Vision Transformer (ViT) with an ensembled regression model. The ViT used is the 'google/vit-base-patch16-224' with pretrained weights. The regression model is an ensemble composed of several base learners, including CatBoost, NeuralNet, RandomForest, and XGBoost, among others. The image encoding derived from the ViT is concatenated with the ancillary data and then used to train the regression model.

Note that the ViT model is only used for inference, and is not trained.

3.3 Training Steps

1. Preprocessing images by applying resizing, cropping and normalization.
2. Apply ViT on the processed images to obtain image encodings.
3. Concatenate image encodings with ancillary data.
4. Feed the concatenated data into the ensembled regression model.
5. Train the model on individual target features to optimize R^2 scores.

The detailed R^2 scores for each target feature on the validation set are provided in Table 3, where 10% of the training set is used as a validation set.

Table 3: R^2 Scores for Each Target Feature Using the Final Model

Target Feature	X4	X11	X18	X26	X50	X3112
R^2 Score	0.4389	0.4217	0.5724	0.2909	0.3019	0.4177

4 Conclusion

The comparative analysis revealed that Vision Transformers outperform traditional CNN architectures in predicting complex plant traits from RGB images. ViT's capability to better capture image features leads to more accurate predictions, as evidenced by higher R^2 scores across various target features.

As shown in Table 1, different transformers yield significantly varying R^2 scores, indicating that each transformer derives unique encodings and extracts distinct features. This observation suggests that an architecture combining the image encodings from multiple transformers, then feeding them into an ensembled regression model, could be worth exploring. With more data, such a model might achieve even lower loss, although this would increase training and inference time and necessitate a much larger regression model.

However, limitations still exist. The current approach relies on the quality of the image and ancillary data, and the effectiveness of preprocessing steps. Future work could involve integrating more sophisticated data augmentation techniques and fine-tuning the Vision Transformer models to further enhance performance. Additionally, investigating other ensemble methods that combine the strengths of both CNNs and ViTs could be worth exploring.

5 Code Availability

Source Code for this report can be found in the following link: ML-Plant-Properties-Predicion

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References

Schiller, C., S. Schmidtlein, C. Boonman, et al. (2021). "Deep learning and citizen science enable automated plant trait predictions from photographs". *Scientific Reports*, vol. 11, no. 1, p. 16395.